

The role of leadership in managing digital transformation: A systematic literature review

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Abstract: Digital transformation should not be seen as a time-bound project, but rather as a process that needs to be continuously improved and managed over time. In the absence of leadership, digital transformation will occur spontaneously in an organisation, but it is necessary to monitor and manage the process itself to achieve the goal and establish a digital organisation. A person who leads the digital transformation must possess the appropriate competencies. Leadership competencies, especially the combination of digital and human skills, are crucial to the development and success of the digital transformation process. The authors of this article identify a vaguely described process of digital transformation as the main research problem, highlighting the need for the systematisation of leadership competencies necessary for effective digital transformation management. In this regard, the research objectives are to define and describe the process of digital transformation and identify the leadership competencies necessary for successful management of the digital transformation process. In order to achieve the objectives of the paper, a systematic literature review was conducted for the period 2001–2022 and 90 papers were identified based on the criteria outlined by the Web of Science and Scopus index databases. The identified papers were then analysed from the aspect of defining digital transformation, the activities of the digital transformation process, and the characteristics of digital leadership. Following a thorough analysis of the selected literature, the paper provides a comprehensive definition of digital transformation, proposes a framework for its management, and analyses the key competencies of digital transformation leaders (digital leaders), which together make up the paper's main contribution. The paper can be used not only as a starting point for future research in this area, but also as initial guidelines for the implementation and management of the digital transformation process. The systematic literature review conducted has certain limitations. Only two, albeit the largest, citation databases were searched, and only papers in English were identified. The established framework should be verified and expanded through empirical research.

Keywords: Digital transformation, leadership, digital leadership, systematic literature review.

JEL Classification: M10, M15.

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Introduction

The existing literature within the research field is insufficiently homogeneous and inadequately integrated, resulting in insufficient knowledge about the competencies required for leaders to successfully carry out the process of digital transformation (Müller et al., 2024). Leaders must develop their individual capacities and competence to provide a better understanding of the change in dynamic environments through a strong global vision (Elidjen et al., 2019). Digital transformation is changing many aspects of how people work (Schwarz Müller et al., 2018), and these changes do not exhibit any signs of deceleration. Leaders have a fundamental role in organisations, such as leading these changes (Evans et al., 2021) and influencing people to perform activities in more efficient and creative ways (Grigore & Coman, 2018). Digital transformation requires leaders to quickly adapt existing approaches in today's highly dynamic environment. Business and IT leaders need to collaborate (since technology and leadership skills are complementary), and new approaches should be adapted to respond to changes in the competitive and technological environment (Hansen et al., 2011; Ko et al., 2022; Philip & Gavrilova Aguilar, 2022). Leadership and digital transformation cannot be considered separately, but they must support each other in the application and sustainability of any digital transformation initiative in an organisation (Jayawardena et al., 2020; Zulu & Khosrowshahi, 2021).

Digital transformation is the driver of positive but also devastating market changes (Al-Nuaimi et al., 2022; Kokolek et al., 2019) and represents more than just a technological evolution (Criado-Perez et al., 2022) that must be adequately managed. As for devastating market changes, Kokolek et al. (2019) argue that technology is increasingly entering everyday life as well as the world of business, and changing how business is conducted. Moreover, emerging IT risks, such as security breaches, privacy concerns, and safety issues (Vial, 2019), alongside implementation costs, can have profoundly adverse effects on the market.

According to Luo et al. (2023), it is necessary to attract highly qualified top managers in order to ensure a successful digital transformation, but they did not address what competencies they should possess. Porfírio et al. (2021) concluded that managers' characteristics, especially democratic leadership styles, are particularly important at the beginning of digital transformation. Furthermore, a three-fold alignment and involvement are necessary, comprising organisation, digital transformation and leadership. According to Carvalho et al. (2023), top managers differ in different industries (sectors or economic activities) and essential changes to the leadership process have been identified based on environmental factors. In this way, a set of characteristics that are essential for the effective achievement of organisational objectives is identified, predominantly from a communication perspective.

As Müller et al. (2024) point out, the literature exists in fragments, and knowledge about the leadership competencies required for digital transformation is lacking. The same authors emphasise that leaders must possess different competencies depending on the drivers and goals of the digital transformation. A key finding of their research is that the classic all-encompassing leadership approach cannot guide digital transformation. Moreover, they assert that different combinations of technical, business, and people-oriented competencies will be needed depending on whether market innovations are explored, operational efficiency is supported, stakeholder involvement is ensured, and competitive positioning is improved.

Thus, digital transformation leadership is becoming increasingly relevant in theoretical and empirical research. The previous statements are illustrated by fragmented and missing knowledge about which leadership competencies the business requires in order to facilitate digital transformation. According to a theoretical perspective, four archetypical competency portfolios are outlined, which are based on the types of leaders who embody the constituent competencies: the challenger (exploring the market innovations), the bricoleur

(supporting operational efficiency), the organiser (ensuring active stakeholder participation), and the competitor (improving competitive position; Müller et al., 2024).

Strategic implications of digital transformation must be considered both at the daily operational level and in the development process. Kringelum et al. (2024) identified five strategic actions that leaders should consider when embarking on digital transformation to succeed with strategic change: i) discussion and communication about vision and strategy; ii) aligning resources and activities with strategy; iii) developing the competencies of top managers, middle managers and employees; iv) ensuring operational improvements and efficiency; and v) creating an orientation toward customer or user needs and expectations.

Based on several sources, Braojos et al. (2024) argue that digital leadership is conceptualised as a style of competencies, behaviours, and practices that inspire and motivate employees in the context of digital transformation. Moreover, Zivkovic (2022) points out that the necessary leadership competencies are flexibility, experimentation, and understanding of digital technologies. In his opinion, digital transformation leaders should not be IT experts, but they should understand how digital technologies impact organisational culture, performance, and development. In support of this statement, Schwarzmüller et al. (2018) state that digital transformation leaders must possess the competence to lead a team of people from diverse cultures. Furthermore, this implies language competence and the ability to lead from a distance as well.

Therefore, digital leadership is crucial for leading digital transformation, i.e., strong leadership is necessary for the success of digital transformation (Zulu & Khosrowshahi, 2021). There is an evident lack of research aimed at understanding digital transformation and leadership itself (Henderikx & Stoffers, 2022), while digital transformation in the digital era requires extraordinary skills and competencies (Noonpakdee et al., 2020).

Kontic and Vidicki (2018) point out that initiative-taking leadership is one of the key factors that will help the organisation become digital. Leadership in the digital era implies a deep understanding of digital business models (Jacobi & Brenner, 2018). Given the new circumstances, Schwaferts and Baldi (2018)

raise the question as to whether standard management concepts are still adequate for the period during and after the digital transformation. Digital transformation must be initiated by the management rather than by the IT department, which could only achieve part of the necessary changes (Ko et al., 2022). Digitisation, digitalisation and digital transformation are the terms often identified with each other (Kohnke, 2017). Digital transformation includes digitisation and digitalisation (Paul et al., 2024), but digital transformation is a more complex and comprehensive concept (O'Leary, 2023). According to Venkatesh et al. (2019), digital transformation has evolved through digitisation and digitalisation up to the digital transformation itself, which encompasses the entire company's business model.

To recognise the necessary characteristics of digital transformation leadership, it is necessary to understand the very concept of digital transformation as well as the implied steps (key activities; Ubiparipovic et al., 2023), as explained in the paper based on the literature. Müller et al. (2024) pointed the theoretical-practical gap in digital transformation management, taking into account the fact that digital transformation requires persistent digital leadership at the top level of the organisation, and that no coherent concept of digital leadership exists today (Müller et al., 2024). Since most organisations are only halfway through the digital transformation process, it is necessary to emphasise that most companies must undertake a challenging path in order to achieve digital maturity. However, 70% of digital transformation initiatives fail to achieve their objectives (Alshehab et al., 2022). The path to digital transformation within an organisation is a multidimensional and complex process that requires effective leadership. This entails that in the digital era, and leaders must adapt their knowledge, skills, and abilities to successfully drive the process of digital transformation within the organisation (Urs et al., 2023).

After reviewing the previous research, a research problem was identified in the form of a vaguely described and defined digital transformation process, along with the need to systematise the leadership competencies required to successfully manage digital transformation. The main objective of the research is to define and describe the process of digital transformation as well as to identify the necessary leadership

competencies for the successful management of the digital transformation process.

To address the main objective of the research, the authors analysed and used 90 selected papers published in the period from 2001 to 2022. A systematic review of the literature was conducted to identify these papers, which is described in the following section. The papers were then analysed from the aspect of three identified sub-units. Based on the analysed papers, a comprehensive definition of digital transformation was proposed, followed by a conceptual framework of digital transformation and key leadership characteristics needed for the successful implementation of the digital transformation process.

1. Research methodology

As part of the research process and the implementation of the systematic literature review methodology, as well as in order to answer the basic research question and achieve the set research goals (which will be discussed in the following sections), three basic segments can be identified (Xiao & Watson, 2019): i) planning the review; ii) conducting the review; and iii) reporting the review (Fig. 1). In accordance with the mentioned methodology and within the first phase of the systematic literature review, the authors managed to: 1) define a specific research problem (general research questions); and 2) develop a complete research algorithm (protocol) to be used for the research in this paper. Step 3 – the initial search within the selected databases by title, abstract and key words was carried out in the second part of conducting the literature review. During this phase, the fourth step entails screening for inclusion by analysing abstracts according to the inclusion criteria. As the research was done in two databases (Web of Science and Scopus, for the period 2001–2022), duplicates were eliminated in the fifth step based on the received list of potential papers. In this segment of the process, the sixth step involves evaluating the quality of the complete papers, as well as respecting the general quality of the papers regarding the methodology applied, concept, content, and results of the research, and in accordance with the inclusion criteria previously mentioned. Once the final set of papers was obtained, an additional review of the literature of the identified papers, as well as papers that cited the identified papers (forward and backward

research) was performed in the seventh step. Finally, in step 8, the analysis and synthesis of the papers were carried out in accordance with the defined concept of the research. A review report was created based on the obtained results in the third segment of the literature review process, which is also the final, ninth step of the comprehensive literature review.

Based on the identified research problem stated in the Introduction section, the following research questions were formulated as the basis of this study in the first stage of the methodology – planning the review:

RQ1: What is implied by the term digital transformation and what are the challenges, factors and steps of the digital transformation process?

RQ2: What leadership competencies and characteristics are necessary to successfully lead the digital transformation process?

Xiao and Watson (2019) also recommend that during certain segments of the literature review process, the research is to be conducted in such a way that the authors are divided into two groups that carry out each step independently, and then integrate their findings and discuss the differences in each step in order to screen the selected papers for final analysis. In Fig. 1, the steps of the applied methodology and the results of the selection as part of the analysis of the papers are described in detail.

Following an initial search and analysis of the literature, the authors identified the insufficient success of the digital transformation process and the inability of businesses to achieve their desired goals through digital platforms. Furthermore, as we delved deeper into the research area, we realised that the key role in digital transformation is played by the human factor, i.e., human versus technology, which led us to include the sub-unit of management or leadership in organisations in our research. A key point that must be noted during this phase of the research is that the research questions were developed through an iterative process, after the authors had gathered more information and knowledge by conducting a literature search and analysis focusing on the most common digital transformation issues.

Fig. 1 shows how the review was conducted according to all stages and relevant criteria and what results each stage reached.

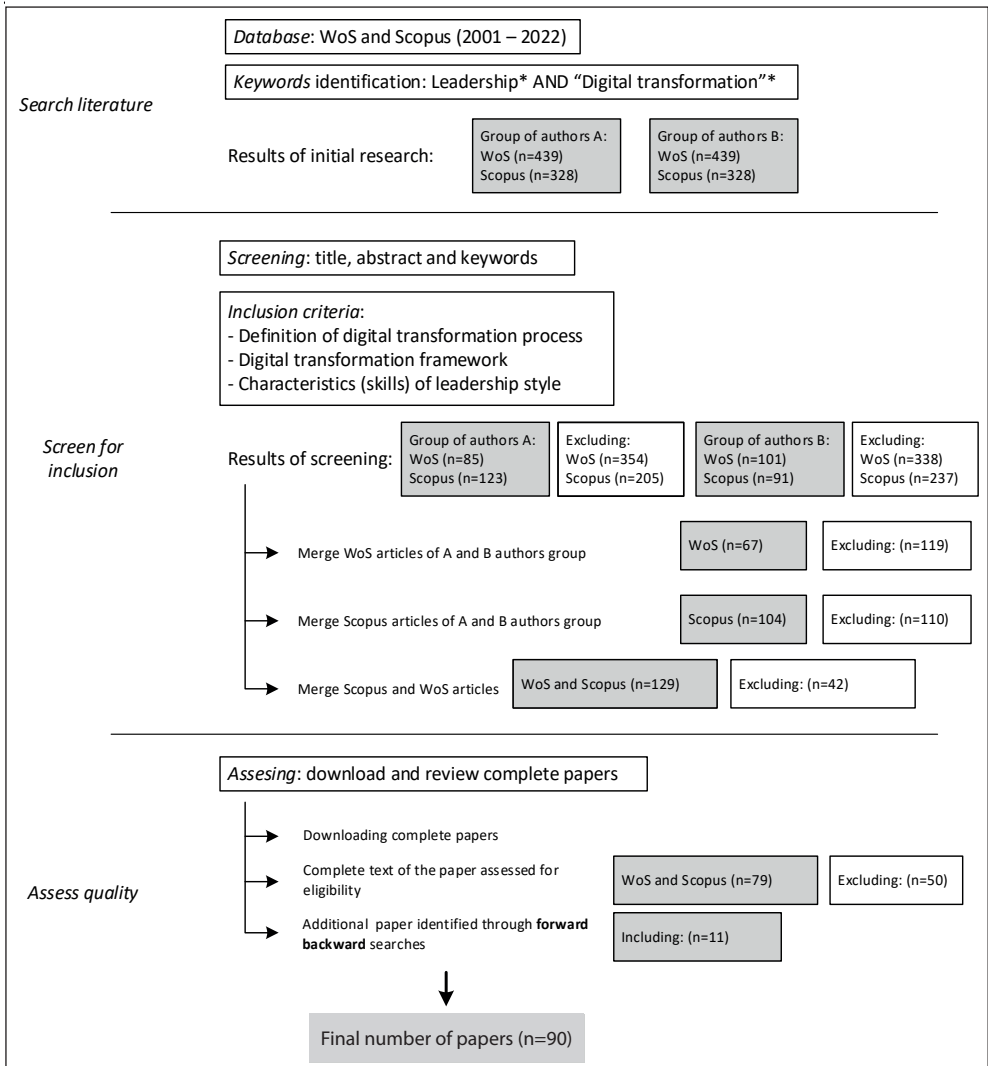


Fig. 1: Search strategy and selection criteria

Source: own

Literature search. Three main search sources were used as literature search channels: i) the mentioned electronic databases (WoS and Scopus); ii) backward searching (identifying the relevant papers cited in the included articles, using the reference list at the end of the paper is a good way to find these articles); and iii) forward searching (finding and selecting articles that cite the original paper that has been found).

The search was performed for the period from 2001 to 2022. This time period was determined based on the topicality of the research area, the volume of published papers and the electronic availability of papers. Based on the research question, the keywords related to “leadership” and “digital transformation” were selected. Among the papers included were only those published in English in journals and

at conferences, as well as chapters in monographs. The search ended after determining that no new results would be obtained by applying the search strategy. Group authors A and B conducted the search within selected databases and defined keywords during December 2022, while the search results, i.e., the number of initial hits, due to the same search criteria, was identical and amounted to WoS $n = 439$ and Scopus $n = 328$.

Screening for inclusion. For screening and selecting papers for final analysis, Xiao and Watson (2019) recommend that at least two researchers evaluate papers independently according to set criteria. Two groups of authors A and B, conducted the literature review separately in this paper, with phase alignment and consultation between both groups. A screening process for selecting papers/including papers was conducted by both groups of authors based on the title, abstract, and keywords of the papers, along with defined inclusion criteria.

Three separate criteria were used to select the papers on which this research is based. On the one hand, these criteria represent the derivative of the research questions, while on the other hand, they serve as the basis for selecting an individual study on which further steps will be taken in the research process. For a paper to be selected for further analysis, it must meet at least one of the following criteria: the first criterion, defining digital transformation – from a leadership position, understanding the digital transformation process is necessary if we are going to ensure its success, which is why we need to define the concept of digital transformation and ensure that its absolute understanding is presented at the beginning of the paper. The second criterion, the conceptual framework, or steps of digital transformation – understanding the essential elements of the concept of digital transformation – is significant for the successful management of the digital transformation process and the achievement of set goals. The third criterion is the leadership characteristics necessary for the successful management of the digital transformation process.

Following the screening procedure previously described, the following results were obtained for the initial list of papers: from the original number of hits ($n = 439$) within the WoS database, the authors of Group A selected

85 papers and excluded 354 papers (Group A, WoS, $n = 85$, excluding $n = 354$), while the authors from the same group within the Scopus database, selected 123 papers and excluded 205 papers (Group A, Scopus, $n = 123$, excluding $n = 205$) from the original number of hits ($n = 328$). From the original number of hits ($n = 439$) within WoS, the authors from Group B selected 101 papers and excluded 338 papers (Group A, WoS, $n = 101$, excluding $n = 338$), while the authors from the same group within the Scopus database, selected 91 papers and excluded 237 papers (Group B, Scopus, $n = 91$, excluding $n = 237$) from the original number of hits ($n = 328$). The next step involved combining the selected papers of the two groups of authors from both databases, removing duplicates, and discussing the reasons for choosing different papers between the two groups of authors. During the discussion, it was decided to exclude papers for which there were not enough arguments for the selection stage. As a result of combining papers from both groups of authors, eliminating duplicates and excluding papers based on discussion within the WoS database, 67 papers remained, while 119 papers were eliminated (WoS, $n = 91$, excluding $n = 237$). By combining the papers of both groups of authors, eliminating duplicates and excluding papers based on the discussion within the Scopus database, 104 papers remained and 110 papers were excluded (WoS, $n = 104$, excluding $n = 110$). The next step of the screening process was to combine papers from both databases and exclude duplicates. After combining the papers from both databases and excluding duplicates, 129 papers remained, while 42 papers were excluded since they appeared in both databases in the previous step (WoS + Scopus, $n = 129$, excluding $n = 42$).

Assess quality. Following the screening process, the next segment of the conducted literature review involved downloading previously selected papers from the screening process and assessing their quality. In preparing the final list of papers for data extraction and synthesis, the quality of the complete papers is assessed. Research questions and the application of defined selection criteria are used to evaluate the papers' completeness. A paper's quality is also assessed based on the method of collecting data, the analysis process, the methodology applied, and

the obtained results and conclusions. Based on the quality assessment, 79 papers were selected and 50 were excluded (preliminary final list, $n = 79$, excluding $n = 50$). During this stage of assessing the quality of the complete papers, the authors from Groups A and B independently assessed each paper in order to harmonise the selected papers and reach the results described previously. As a final step toward reaching the final list of papers which will serve as the basis for this paper, forward and backward searches were conducted, and an additional 19 papers were included on that basis (for and back search, $n = 11$), bringing the total number of papers selected to be 90 to provide a response to the research question (final number of papers, $n = 90$).

Extract data. Based on the literature review of the selected papers, information was extracted on two topics: i) definition of digital transformation and conceptual framework of digital transformation; and ii) characteristics, role and style of leadership. Data extraction from the selected papers is guided by the method of synthesis selected, which, in this case, is a textual narrative synthesis. Generally, coding is used to extract data, especially during a literature review, which was conducted inductively according to selected topics.

Analyse and synthesise data. After the data was extracted, the authors organised the data in accordance with the review concept. In this phase of the literature review, techniques such as tables and text descriptions were used. The method of contingent design was used as the main method of analysis and synthesis of the extracted data, which represents a cyclical research synthesis. Consequently, the selected quantitative and qualitative papers are intended to address a specific research question.

2. Research findings

2.1 Defining digital transformation

The continuous emergence of new technologies is evident in recent years (Zhu et al., 2021). Although digital transformation has penetrated many aspects of organisations, transformed business models (Danielsen, 2020), in some cases disrupting existing businesses and economies (Philip & Gavrilova Aguilar, 2022; Zimmermann et al., 2019), being inevitable in all industries (Gurkan et al., 2020), and becoming a buzzword in the last few years (Abbu et al., 2021), it is a relatively new concept

that is very difficult to define clearly (Kokolek et al., 2019). The authors generally agree that digital transformation is a process (Egor, 2020; Engesmo & Panteli, 2021; Lorentzen, 2022) that manifests itself in different ways and with different intensity (Engesmo & Panteli, 2021). As digital transformation is a continuous process, its monitoring and management are necessary (Egor, 2020). It can be said that digital transformation is an inevitable and irreversible process (Zhu et al., 2021) that represents a chance for innovation within organisations (Zulu & Khosrowshahi, 2021) driven by clients' expectations and behaviour (Evans et al., 2021). Digital transformation is often associated with technology, corporate culture, society, creativity and change management (Tungpan-tong et al., 2021). Many organisations focus digital transformation too much on technology alone. This is precisely why many digital transformation processes fail (Jayawardena et al., 2020). Digital tools cannot just be purchased and simply attached to an organisation (Egor, 2020). Needless to say, digital technologies cannot be neglected in the digital transformation, because they play one of the main roles in the digital transformation (Vial, 2019). A company should learn about new technologies, as well as develop new capabilities that allow it to reposition itself strategically, because new technologies can cause disruptions that can threaten the business of a particular company (Pihlajamaa et al., 2021). Therefore, it is impossible to talk about digital transformation without mentioning the technologies, in which the authors often include, e.g., social media, internet of things, mobility, big data and analytics, cloud computing, artificial intelligence and blockchain (Florek-Paszkowska et al., 2021; Tekic & Koroteev, 2019). From these groups of technologies, it is necessary to choose and apply the ones (Venkatesh et al., 2019) that are most appropriate to the organisation or to the strategy of digital transformation.

Carcary et al. (2016) even state that organisations should not initially focus on a holistic set of required capabilities, but on the development of basic capabilities that will become the starting point for digital transformation and later lead to the development of all necessary digital capabilities. The same authors point out the following as basic or key areas on which organisations should initially focus: promoting and implementing an agile digital culture, developing

digital leadership skills, building digital talent and defining and implementing a transformative digital business strategy. According to Zhu et al. (2021), digital transformation implies a transition from a product-centric to a customer-centric strategy. Therefore, the management needs to adapt its business strategy to the digital reality (Reis & Melao, 2023).

Florek-Paszowska et al. (2021) group the levers that will influence successful digital transformation into two groups: human-based and non-human based factors. The first group includes, in the first place, competent, responsible, and open-minded leaders, who are open

to change, flexible and able to quickly adapt. The authors then mention talented employees who are ready to share their expertise, learn quickly, and adapt, as part of the first group. The second group, non-human based factors, implies a business culture that should empower employees, and business models that are easily transformed in order to create an innovative business model, new strategies and support (disruptive) technologies. This can be followed by team-based levers that enable digital transformation, which were identified in their research by Guinan et al. (2019): diverse team composition, iterative goal setting, continuous learning

Tab. 1: The definitions of digital transformation

Source	Definition/explanation of the concept of digital transformation
Reis and Melao (2023)	Digital transformation is the use of new digital technologies that enables major business improvements and influences all aspects of customers' lives.
Kutnjak et al. (2019)	Digital transformation is a complex and demanding process that requires the engagement of all enterprise resources – human, technological, physical, organisational and financial. DT is the transformation of an organisation driven by digital technology, and it directly affects the change of the organisation's core, that is, the change of its business model.
Brown and Brown (2019)	Digital transformation is defined as the implementation of innovative and new digital technologies to effect business improvements in an organisation.
Venkatesh et al. (2019)	Digital transformation or digital business transformation is thus about embracing digitisation and digitalisation concepts to create novel and different business designs or business model as a whole for better long-term value. It is much more than automating or digitising business processes or digitising products and services.
Vial (2019)	Digital transformation is a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies.
Tu and Chi (2020)	Digital transformation has been defined as the use of new digital technologies, such as mobile, artificial intelligence (AI), cloud, blockchain, and IoT technologies, to enable major business improvements to augment customer experience, streamline operations, or create new business models. The key long-term technological trends are the increasing capability of the firm with customers to communicate, collect, store and analyse information.
Fachrunnisa et al. (2020)	Digital transformation is a process or business for the company in facilitating the relationship between customers with the company itself, simplifying the various processes by changing the business model through the recent technology. This change is not only limited to the use of technology but also has an impact on the structure and strategy of the company to fit the business model due to the new technology.
Abdallah et al. (2021)	Digital transformation (DT) is a customer-centric mechanism that enables continuous improvement in the productivity of the manufacturing processes using advanced digital technologies such as cloud computing, internet of things, big data analytics, digital twin in all aspects of the organisation.

Source: own

and talent management. Based on the research conducted, Engesmo and Panteli (2021) suggest that organisations should adapt their IT functions to support digital transformation. When discussing digital transformation, it is important to take into account that digital business models often require the organisation to participate in or initiate a certain digital ecosystem in which traditional relationships with suppliers and customers are challenged (Criado-Perez et al., 2022; Schwaferts & Baldi, 2018).

Reis and Melao (2023) identified three different elements of digital transformation based on a review of different definitions of digital transformation. The first element includes the technological digital transformation that encompasses technologies (e.g., social media, mobile, analytics). The second element is the structural digital transformation, which requires changes in business processes and the creation of new business models. Finally, as the third element, the authors mention the human digital transformation that affects all aspects of human life (e.g., improving customer experience). According to Weritz et al. (2020) three main parts of digital transformation are products (services), clients and processes. Tab. 1 shows the definitions of digital transformation identified through a systematic review of the literature.

2.2 The framework of digital transformation

The road map for digital transformation should be developed, which will guide the organisation to where it wants to be in the future (Evans et al., 2021). The problem is that digital initiatives are not always strategically oriented and clearly defined (Engesmo & Panteli, 2021), or management often does not see the difference

between digital transformation initiatives and smaller-scale initiatives aimed at increasing efficiency (Solberg et al., 2020). In other words, a non-strategic view of digitisation is precisely what is reflected here.

Kokolek et al. (2019) state that the prerequisite for the success of digital transformation is the acquisition and development of six key factors, as follows: digital knowledge and skills, ability to apply new technologies to the business, continuous learning, strategic vision, network leadership and agility. Criado-Perez et al. (2022) point out that traditional organisations cannot adapt to the changing environment by solely relying on educational institutions as they once did. They argue that employee training is essential for organisations to adapt to the changing environment. As challenges of digital transformation, Hai et al. (2021) cite limited infrastructure, lack of skills and the fact that management has not adapted to the digital transformation process. Digital transformation is a widely adopted process, which significantly complicates its successful implementation. Therefore, companies need both employees who are engaged and support the digital transformation process, as well as leaders who are reliable and possess adequate competencies to lead and support employees through the digital transformation process (Heracleous & Gledhill, 2024). As a result, digital transformation presents a number of challenges to organisations (Tab. 2 provides the most significant challenges).

Imran et al. (2021) identify leadership, organisational structure, and culture as the main drivers of digital transformation. According to their research, developing leadership competencies facilitates digital transformation

Tab. 2: Challenges of digital transformation

Challenge	Source
Limited infrastructure	Hai et al. (2021)
Lack of skills	Hai et al. (2021)
Not adapted management, lack of knowledge to lead digital transformation	Hai et al. (2021), Salume et al. (2021)
Lack of resources to incorporate digital technologies	Van Tonder et al. (2020)
Culture in an organisation	Evans et al. (2021)
Employee resistance	Trenerry et al. (2021)

Source: own

because organisations become more agile, more customer-centric and more collaborative. They mention organisational structure and culture as the next drivers. Having a flexible organisational structure and an organisational culture that embraces change will make implementing digital transformation easier.

Digital transformation requires commitment from management, while the IT department plays a less significant role (Ko et al., 2022). According to Ko et al. (2022), a strategic framework for digital transformation is one of the key activities of digital transformation. Although digital transformation is necessary for the survival of modern organisations (Trenerry et al., 2021), the management often lacks adequate knowledge about the different elements that should be considered when transforming a business digitally (Salume et al., 2021). Organisations

today face a big challenge when it comes to managing digital transformation (Boneva, 2018). As a result, identifying the key factors of digital transformation is vital (Tab. 3).

Strategy in the digital age must be swift and supported by a roadmap for implementation (Jacobi & Brenner, 2018). As a result, digital transformation is not much different from other business transformations, and a roadmap is required (Evans et al., 2021; Jacobi & Brenner, 2018; Kohnke, 2017; Zaoui & Souissi, 2020). Organisations should identify the steps they should follow in their digital transformation process. Tab. 4 shows the most frequently mentioned steps of digital transformation in the literature. It is important to consider that they are not sequential and should be implemented according to the needs of the industry and organisation on the path to digital transformation.

Tab. 3: Key factors of digital transformation

Factors	Source
Customer centricity	Mhlungu et al. (2019), Abdallah et al. (2021)
Corporate processes and structure	Imran et al. (2021), Jacobi and Brenner (2018), Alshehab et al. (2022)
Digital knowledge and skills	Kokolek et al. (2019), Imran et al. (2021)
Agility	Kokolek et al. (2019)
Leadership	Kokolek et al. (2019), Imran et al. (2021), Jacobi and Brenner (2018), Alshehab et al. (2022), Trenerry et al. (2021)
Continuous learning/upgrading/retraining	Kokolek et al. (2019), Weritz et al. (2020)
Commitment of the management	Ko et al. (2022)
Strategic vision, creation of the strategic framework of DT	Kokolek et al. (2019), Ko et al. (2022)
Ability to apply new technologies to the business	Kokolek et al. (2019),
Possessing the necessary capabilities to ensure digital transformation	Van Tonder et al. (2020)
Creating a digital transformation team	Evans et al. (2021)
Culture and people	Jacobi and Brenner (2018)

Source: own

2.3 Leadership in process of digital transformation

Based on a systematic review of the literature, we found that three aspects are present within the analysed publications that explicitly discuss

the definition of digital leadership. Firstly, existing literature does not clearly define the relationship between e-leadership and digital leadership. E-leadership is defined as a process of social influence with the mediation

Tab. 4: Steps/dimensions of digital transformation

Steps/dimensions of digital transformation	Van Tonder et al. (2020)	Gaffney and Pelser (2021)	Evans et al. (2021)	Jacobi and Brenner (2018)	Peter et al. (2020)	Andriole (2020)
Install digital leadership				✓	✓	✓
Development of new products/services and/or digitalisation of existing ones	✓					
Development of employees' digital competences		✓		✓		
Digitalisation of customer platforms (channels)	✓					
Transforming the user experience (customers, partners, workforce)						
Identifying and acquiring the necessary resources for incorporating digital technologies	✓					
Digitalisation of business processes					✓	✓
Changing/modifying the organisation – structure, incentives and culture			✓			✓
Digitalisation of existing capabilities	✓					
Building a digital strategy	✓	✓		✓	✓	
Creating cross-functional teams		✓		✓		
Building stronger ties with stakeholders				✓		
Collecting digital knowledge and experience at the organisational level				✓		
Creation and measurement of key performance areas KPAs		✓				

Source: own

of advanced information technologies in order to change the attitudes, feelings, opinions, behaviour and performance of individuals, groups and organisations. Furthermore, e-leadership does not change the basis of business; rather, technology helps to realise business. Also, some authors treat e-leadership and digital leadership as synonyms. Secondly, the analysed literature provides a concise definition of digital leadership that blurs the distinction between e-leadership and digital leadership. Thirdly, the results show that e-leadership and digital leadership are not synonyms, suggesting that digital leadership is much broader than e-leadership, as e-leadership primarily uses technology to support existing business. Digital leadership is an important component

of achieving the goal of digitally enabled business model, digital organisations, and management of employees (Eberl & Drews, 2021).

Digital leadership entails the ability to direct resources and utilise information communication technologies to achieve the organisation's goals, significantly contributing to organisational performance (Suryadi Muslim & Setyono, 2024). Furthermore, the leadership required to achieve digital transformation can be identified by the strategic leadership of digitalisation, which involves leaders' ability to create a clear and understandable vision for the digitalisation process as well as the ability to implement the strategy in order to achieve the digital transformation (Engesmo & Panteli, 2021). Digital leadership is a combination of a transformational leadership

style and the use of digital technology (Mihardjo & Rukmana, 2019), or even a combination of digital competencies and digital culture (Mihardjo et al., 2019). Digital leadership can also be defined as the complete transformation of business operations, as well as the enhancement of organisational performance in specific areas through the application of technology across all aspects of business (Arabiun et al., 2024).

Values, skills and behaviour of leaders are crucial in directing and leading organisational development processes (Zivkovic, 2022).

Taking into account the role of leadership in the process of digital transformation, starting with authors that emphasise the exclusive responsibility for digital transformation success lies with top management (Abbu et al., 2022; Kontic & Vidicki, 2018; McCarthy et al., 2021; Sow & Aborbie, 2018), the research results indicate that digitally mature organisations support agile principles to a much greater extent such as transparent and open communication regarding digital projects and draw and generate new ideas based on communication

Tab. 5: Competencies/skills/characteristics of digital leaders

Source	Characteristics of leadership
Karakose et al. (2021)	Use of technology, managerial skills, individual skills
Zivkovic (2022)	Vision, innovation, flexibility, understanding of digital technologies, collaboration, empowerment, multiple intelligences, continuous learning, experimentation
Guzmán et al. (2020)	Cognitive skills, interpersonal skills, strategic skills
Gilli et al. (2023)	Cooperation, strategic thinking, team leadership skills, customer orientation, communication skills
Imran et al. (2021)	Adaptability, data-driven decision-making, understanding new digital technologies, risk taking ability, vision, empowerment, failing fast
Schiuma et al. (2022)	Mentor, pragmatist, catalyst, communicator, righteous, facilitator
Schiuma et al. (2021)	Righteous, dreamer, builder, charismatic, bright, generous
Abdallah et al. (2021)	Flexibility, diversified knowledge, priority and results focus, ownership and responsibility
Bygstad et al. (2017)	Deep employee participation, leadership that leverages employee knowledge
Cortellazzo et al. (2019)	Digital skills, human skills, change initiation, problem solving
Cuya and Bayona-Oré (2022)	Communication knowledge, technological knowledge
Eberl and Drews (2021)	Visionary, digital literacy, cooperation, adaptability, motivation
Imran et al. (2021)	Digital vision, digital knowledge, empowerment, management of diverse teams, risk taking
Larjovuori et al. (2018)	Strong vision and clear goals, commitment and investment, leading cultural changes, coaching, promoting participation, customer orientation, collaboration and partnership
McCarthy et al. (2021)	Digital architect, digital strategist, digital culturalist, customer centrist, business process optimiser, digital workplace landscaper, data advocate, organisational agilist
Mihardjo et al. (2019)	Communication, computing, content, telecommunications
Noonpakdee et al. (2020)	Business, digital technology, leadership, process, change (strategy)

Source: own

with clients, suppliers and business partners to a much greater extent. Digitally mature organisations also have leaders who have a high level of education, exhibit entrepreneurial behaviour, and foster digital transformation. In conclusion, leaders in digitally mature organisations contribute significantly to digital transformation. Overcoming negative perceptions of organisational digital transformation is possible by adopting new trends in developing the skills of employees, especially leaders (Sousa & Rocha, 2019). The successful leader in the digital age is one who has leadership skills (Mihardjo et al., 2019), a mindset that adapts to digital transformation, and is able to recognise and seize the opportunity to be informed, make timely decisions in order to adjust strategy when risky situations arise, contributing significantly to the organisation's optimal development (Hai et al., 2021).

Tab. 5 presents a list of the most common competencies of digital leadership that are cited in the analysed papers as research results. However, a conclusion can be found that there is no single conceptual framework of digital leadership competencies, which are not unified (Wahyuningsih & Asri, 2024).

According to previous studies, digitally mature organisations differ from digitally developing organisations when it comes to leadership competencies. As a result, leadership competencies play an important role in the development and success of the digital transformation process, particularly the combination of digital and human skills (Cortellazzo et al., 2019). One of the key competencies of a leader in the digital era is trust (Savolainen, 2018), which can be the basis of a separate leadership style. Digital transformation requires a new form of leadership (Kohnke, 2017), while innovations in digital leadership are set as a new type of support in the form of digital innovations (Nurhidayati & Ratnasari, 2021; Wasono & Furinto, 2018) of the business model (Yopan et al., 2022). In order to achieve digital leadership, there are three leadership styles – administrative leadership, adaptive leadership, and enabling leadership, on the one hand, and four platforms – an adoption platform, a learning platform, an innovation platform, and a system and business platform – on the other (Tanniru et al., 2018).

Digital leadership enabled business model by: i) transforming the role, skills, and style

of the digital leader; ii) realising a digital organisation, including governance, vision, values, structure, culture, and decision processes; and iii) adjusting management, virtual teams, knowledge, and communication and collaboration on the individual level (Eberl & Drews, 2021). In the digital age, leadership is clearly becoming a recognised leadership style based on two dimensions that are balanced. Firstly, it is characterised by a behavioural perspective and a focus on the human aspects and teamwork of the organisation, but on the other hand, it is characterised by a strong focus on innovation and digital technology adoption (Oberer & Erkollar, 2018; Weber et al., 2022), which is affected by digital transformation (Panshin et al., 2021).

Leaders in today's organisations need to ask a few key questions (Ko et al., 2022): How will the development of digital technologies and business models affect organisations in the future? How well do we understand the power of digitalisation that determines our present and are we capable of making effective choices about how to change it? How to lead the transformation of the company, and develop the necessary organisational capabilities?

Successful digital transformation begins with a change in mindset among employees, leadership, and throughout all levels of the organisation. These changes lead to a cultural shift that enables the organisation to be agile and ready to take risks. Organisations that initiate this process with a focus solely on technology often find themselves disappointed with the results (Kane, 2019). It is also important to highlight that ineffective leadership is the primary cause of digital transformation failure (Aditya et al., 2021). In the information systems research, top management support is often regarded as a positive phenomenon essential to successful digital transformation, including the implementation of information systems associated with organisational, operational, and strategic changes (Alieva & Powell, 2023; Elbanna & Newman, 2022; Gudergan et al., 2021; Gurkan et al., 2020; Wrede et al., 2020).

Conclusions and discussion

Organisations today are more or less affected by digital transformation. Generally, the authors agree that digital transformation is not a one-time project, but rather a process that manifests itself in different ways and at different

intensities. Without management, digital transformation will happen spontaneously in the organisation, but to achieve the desired results, it must be monitored and managed. There is no doubt that digital transformation is technologically based, but the emphasis should not be solely on technology. Digital transformation is often associated with the following technologies: social media, the internet of things, mobility, big data and analytics, cloud computing, artificial intelligence and blockchain. The main factors enabling digital transformation include (among others) technologies. A customer-centric approach and an effective digital strategy that includes, among other things, the digital ecosystem, or, extending beyond the traditional boundaries of the organisation are essential.

The literature does not offer a generic and generally accepted definition of digital strategy. This is due to the fact that digital transformation takes a different form in different organisations, and it is viewed differently by experts. The following characteristics can be attributed to digital transformation according to the literature review: digital transformation is a continuous process of changing business processes; digital transformation includes the creation of new or the modification of existing business models; digital transformation is more than the implementation of new technologies;

digital transformation includes new digital technologies that enable improvements and/or transformation of business processes; digital transformation is a customer-centric mechanism that enables continuous improvements; digital transformation includes the following technologies among others: social media, internet of things, mobility, big data and analytics, cloud computing, artificial intelligence, blockchain, and cloud computing.

Despite the fact that digital transformation can take place spontaneously within organisations, it needs to be managed professionally. The management must have well-defined paths, or road maps, for leading the digital transformation effectively. A strategic approach to digital transformation is very important. Digital transformation challenges must first be identified within the organisation. Limited infrastructure, a lack of digital skills among employees, management's inability to adapt, a lack of resources to integrate digital transformation, organisational culture, a lack of clear plan, and employee resistance all contribute to the challenges of digital transformation. It is essential to identify the challenges of digital transformation, take measures to overcome them, and analyse the key factors of digital transformation. There are numerous actions that must be carried out simultaneously, starting with

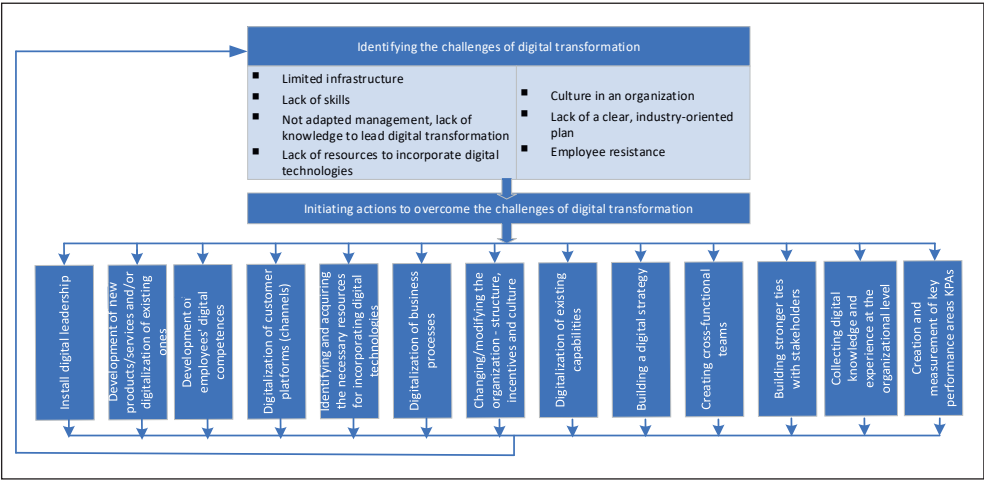


Fig. 2: Framework of digital transformation

Source: own

developing a digital strategy, installing digital leadership, and creating and measuring key performance areas (Fig. 2).

All of the above supports the position that leadership (people, strategy, leadership, business processes and technology) is the fundamental component of digital transformation (Abbu et al., 2022; Abdallah et al., 2021; Grigore & Coman, 2018). Leadership and digital transformation are closely related, which starts with a leader with possessing visionary thinking, agility, an understanding of the value of data, data-driven decision-making, knowledge of strategy, and a willingness to embrace change and changes in culture and networking (Larjovuori et al., 2018; Philip et al., 2023).

The readiness of an organisation for transformation can be determined by a number of factors. From the position of practical and academic views, it is concluded that organisations in this process require not only individual commitments, such as executive directors, but also teams of competent managers, who will take responsibility for the success of the digital transformation process. Whether they succeed or fail depends greatly on their ability to build leadership trust and apply it for transformation. Digital transformation research suggests that leaders create a roadmap for the transformation process so as to know where they want their organisations to be in three to five years. A chief executive officer (CEO) plays a very important role in the transformation process, and managers and leaders will face significant challenges in the future (Cortellazzo et al., 2019; Zeike et al., 2019). Changing the way an organisation thinks has a big impact on leadership and management (Evans et al., 2021).

Digital transformation brings significant challenges when it comes to managerial competencies. In order to improve business efficiency and introduce digital innovations, digital technologies require new competencies that must be combined with existing ones (Henderikx & Stoffers, 2022; Pihlajamaa et al., 2021), since the development of higher levels of digital transformation is far more related to the characteristics of the management than to the characteristics of the organisation (Porfirio et al., 2021).

A key requirement for the positive development of the change process is a change in leadership style, which managers realise is necessary. Study results indicate that managers need to remove themselves from their

personal comfort zones in order to learn a new way of behaving and implement changes. To address possible uncertainties (and potential fears) on the part of employees, managers believe it will be helpful to define a clear vision that will enable the new culture of leadership to flourish (Sokolovic et al., 2022). On the other hand, they propose specific goals that can be achieved with AI in leadership and culture and indicate the personal use of new technologies.

Contribution, limitations and suggestions for future research. Both practitioners and researchers can benefit from a clearer definition of digital transformation. This proposed framework can serve as a starting point for digital transformation research, both for researchers and for organisations managing digital transformation. In establishing training requirements and recruiting adequate employees, HR management can utilise the proposed competencies list. There are certain limitations in the conducted systematic literature review. Only two, albeit the largest, citation databases were searched, and only papers in English were identified. The established framework should be verified and expanded through empirical research (Gudergan et al., 2021), due to a lack of empirical research that addresses the relationship between digital leadership and digital transformation process (Müller et al., 2024). Both HR management and experts who are familiar with and lead digital transformations in organisations should empirically verify the identified competencies. Future research should focus on the relationship between digital leadership and organisational performance (Abbu et al., 2022), as well as the development of the digital maturity assessment model, which comprises a variety of dimensions, although most of them cover technology, people, and strategy. Furthermore, future research should also analyse companies with a high level of digital transformation maturity and identify the characteristics of leaders that enabled this maturity and the degree of implementation of the digital workplace.

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